



20TB storage system

Western Digital announced a new 20TB version of its My Book Duo storage system at \$849 (roughly 55,000 INR). <http://dgit.in/20TBmyBook>



Discord vs Twitch app

Discord has new video chat and screen sharing features, meanwhile, Twitch just officially launched their new App. <http://dgit.in/DisTwitch>

Are you even listening?

O

ur ears play a major part in how we perceive and react to our reality. For instance, an approaching siren always makes you step aside from the road and a high pitched shriek will heighten your senses. These instinctive reactions, coupled with how our brain perceives different type of sounds, leads to a lot of sound illusions. These illusions are particularly used by sound engineers, be it for movies, music or games. Let's have a look at some interesting examples.

P.S Familiarity with some audio jargon will help you a lot with this one

BOOM!

When the ear experiences loud noises, the middle ear muscles contract to keep the delicate cochlea safe from the increased vibrational energy

being transmitted. As a result, the loudness reduces immediately after the initial burst. However, the brain is used to sensing this sudden change and still perceiving the sound as loud and sustained.

This particular trick is used in cinematic sound design to create the illusion of sustained loud noises, like explosions and gunfire, without actually giving a theatre full of moviegoers hearing damage. It's also used by music producers, especially

to punctuate the drops in a club or electronic track.

SHEPHERD'S PARADOX

This particular illusion is very interesting. A pair of chords being played sound as if they are continuously rising up in scales, but in fact they get nowhere. In fact, if you loop a sample like this, it will be impossible to detect where it starts and where it ends.

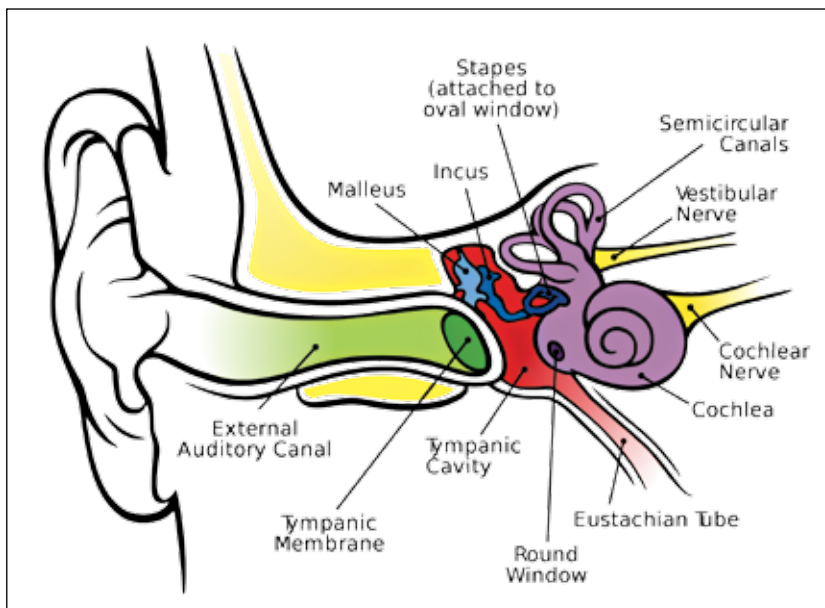
If you're wondering where you might have heard a tone like this, you need to look no further than the sound of the batpod in *Dark Knight* and *Dark Knight Rises*. Since the filmmakers didn't want the batpod to change gears and continuously accelerate, a Shepherd tone was used to create the special sound of Batman's motorcycle-esque vehicle. You should also check out how it was used to create an increasing sense of danger more recently in *Dunkirk*, another Christopher Nolan movie, here: <http://dgit.in/Nolanoise>.

THE HAAS EFFECT

Helmut Haas described this effect in 1949. He was studying the relation between original sound and its initial reflections with respect to human hearing. According to his research, as long as the initial reflections are no more than 35ms delayed and 10dB

Sound is a tricky thing,
and it is pretty easy to fool
our ears

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A helpful diagram of the ear that you might need for some of the concepts in this article